

Classification Error Metrics

Choosing the Right Error Measurement

- You are asked to build a classifier for leukemia
- **Training data:** 1% patients with leukemia, 99% healthy
- **Measure accuracy:** total % of predictions that are correct

Choosing the Right Error Measurement

- You are asked to build a classifier for leukemia
- **Training data:** 1% patients with leukemia, 99% healthy
- **Measure accuracy:** total % of predictions that are correct
- Build a simple model that always predicts "healthy"
- Accuracy will be 99%...

Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Confusion Matrix

	Predicted Positive	Predicted Negative	
Actual Positive	True Positive (TP)	False Negative (FN)	← Type II Error
Actual Negative	False Positive (FP)	True Negative (TN)	

↑
Type I Error

The diagram illustrates a 2x2 confusion matrix. The columns represent predicted outcomes (Positive and Negative) and the rows represent actual outcomes (Positive and Negative). The cells are color-coded: blue for correct classifications (TP and TN) and red for misclassifications (FN and FP). An arrow points from the 'Type II Error' label to the False Negative (FN) cell. Another arrow points from the 'Type I Error' label to the False Positive (FP) cell.

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Accuracy: Predicting Correctly

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}}$$

Recall: Identifying All Positive Instances

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

$$\text{Recall or Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Precision: Identifying Only Positive Instances

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Specificity: Avoiding False Alarms

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

$$\text{Specificity} = \frac{\text{TN}}{\text{FP} + \text{TN}}$$

Error Measurements

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Error Measurements

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall or Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{FP} + \text{TN}}$$

Error Measurements

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}}$$

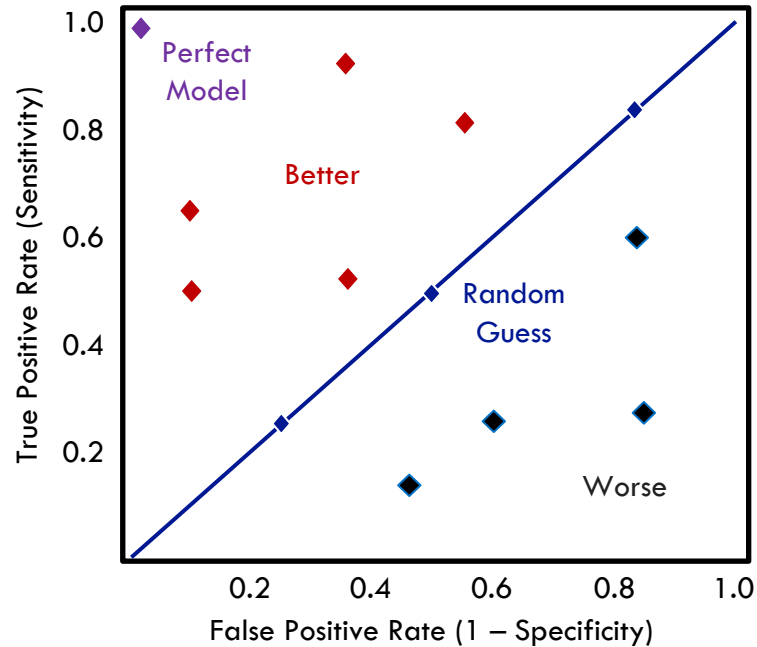
$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall or Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{FP} + \text{TN}}$$

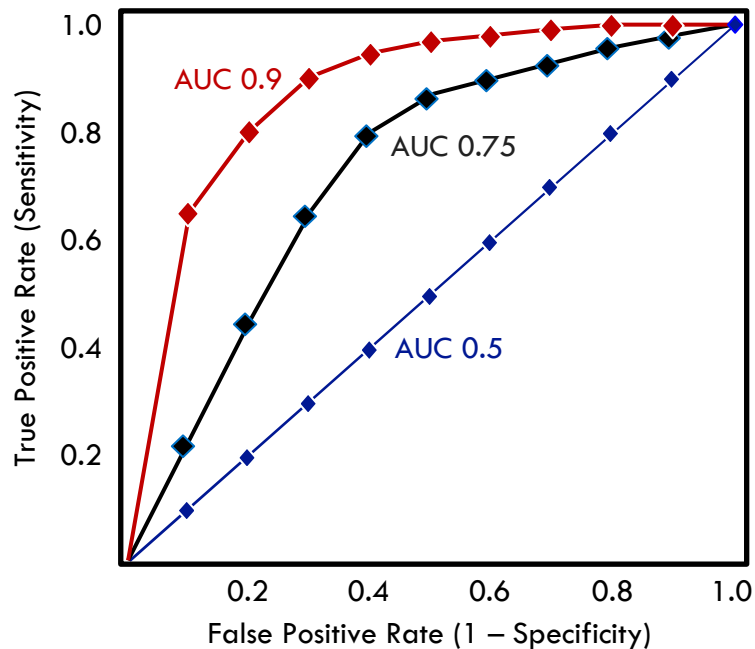
$$\text{F1} = 2 \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

Receiver Operating Characteristic (ROC)



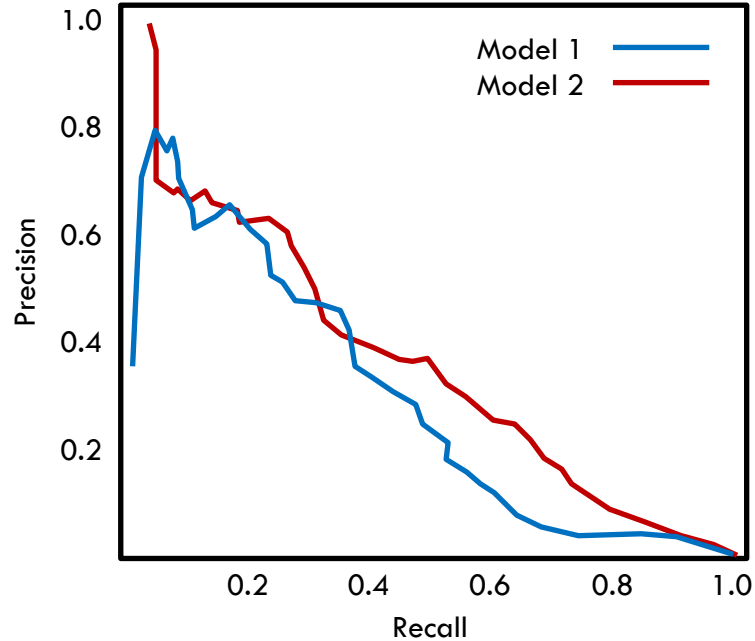
Evaluation of model at all possible thresholds

Area Under Curve (AUC)



Measures total area under ROC curve

Precision Recall Curve (PR Curve)



Measures trade-off between precision and recall

Multiple Class Error Metrics

	Predicted Class 1	Predicted Class 2	Predicted Class 3
Actual Class 1	TP1		
Actual Class 2		TP2	
Actual Class 3			TP3

Multiple Class Error Metrics

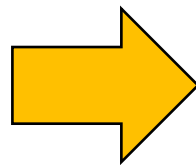
	Predicted Class 1	Predicted Class 2	Predicted Class 3
Actual Class 1	TP1		
Actual Class 2		TP2	
Actual Class 3			TP3

$$\text{Accuracy} = \frac{\text{TP1} + \text{TP2} + \text{TP3}}{\text{Total}}$$

Multiple Class Error Metrics

	Predicted Class 1	Predicted Class 2	Predicted Class 3
Actual Class 1	TP1		
Actual Class 2		TP2	
Actual Class 3			TP3

$$\text{Accuracy} = \frac{\text{TP1} + \text{TP2} + \text{TP3}}{\text{Total}}$$



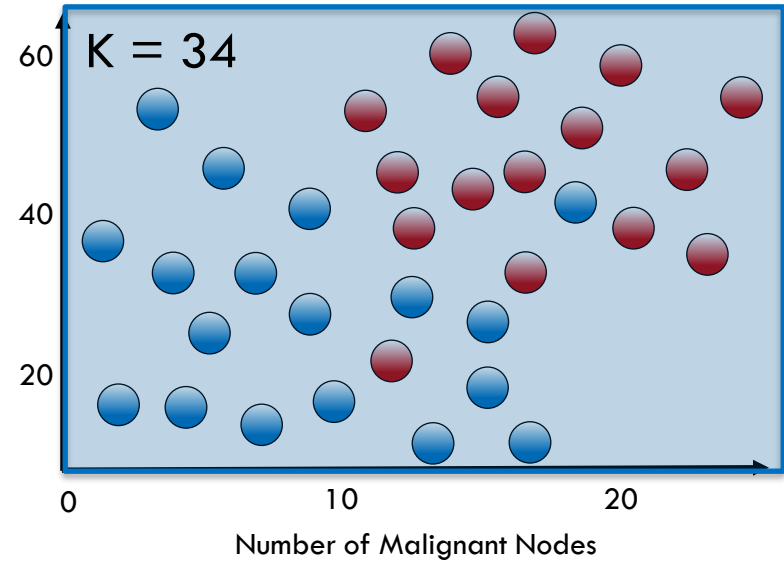
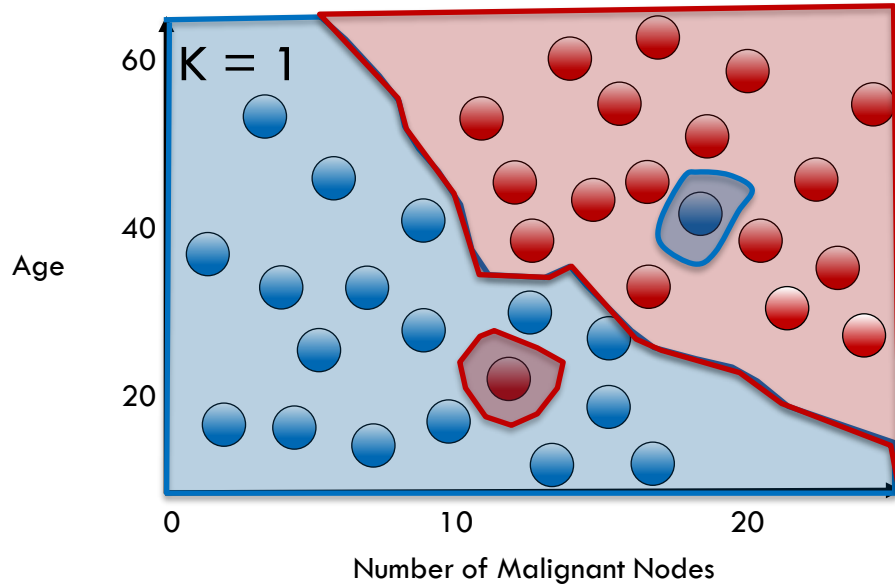
Most multi-class error metrics are similar to binary versions—just expand elements as a sum

Model Generalization

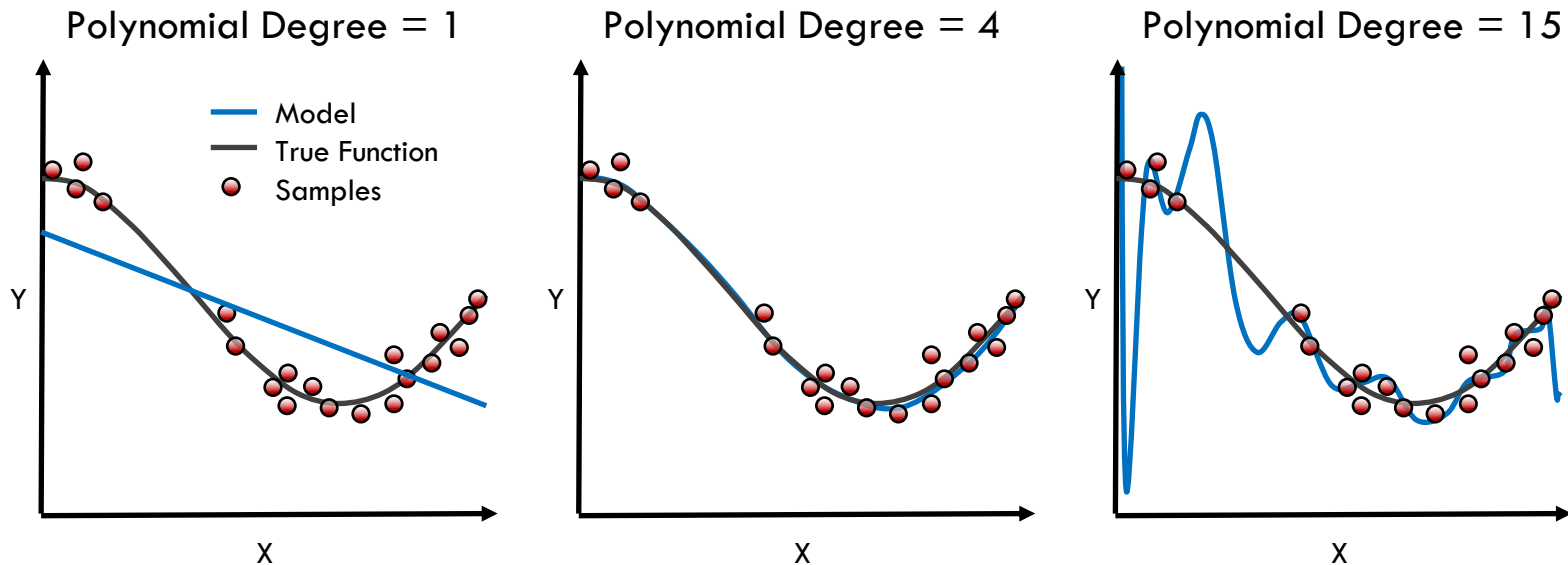
Learning Objectives

- Explain the difference between over-fitting and under-fitting a model
- Describe Bias-variance tradeoffs
- Find the optimal training and test data set splits, cross-validation, and model complexity versus error
- Apply a linear regression model for supervised learning

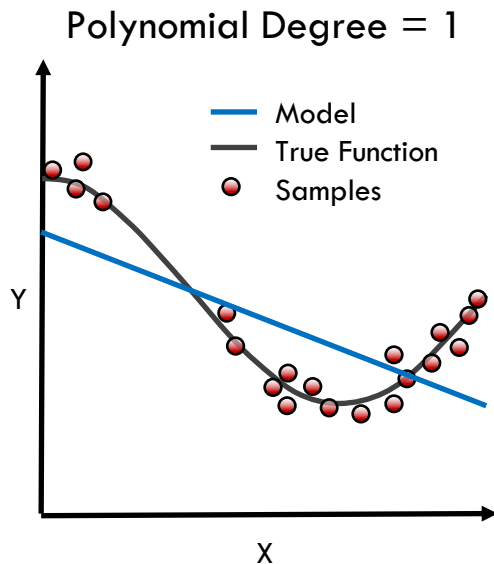
K Value Affects Decision Boundary



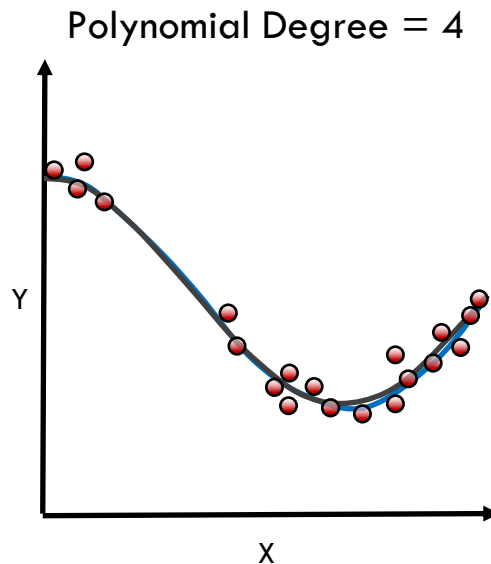
Choosing Between Different Complexities



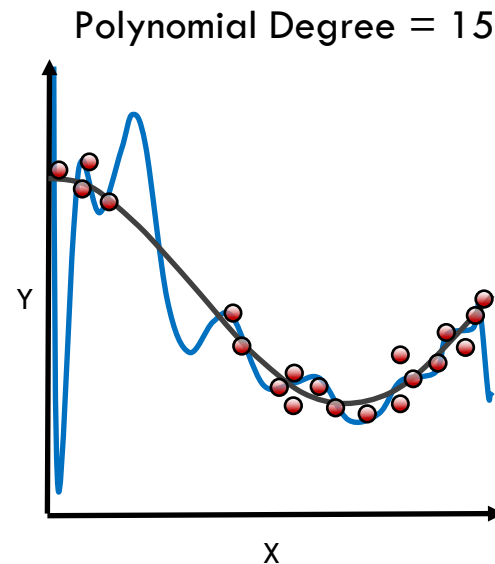
How Well Does the Model Generalize?



Poor at Training
Poor at Predicting

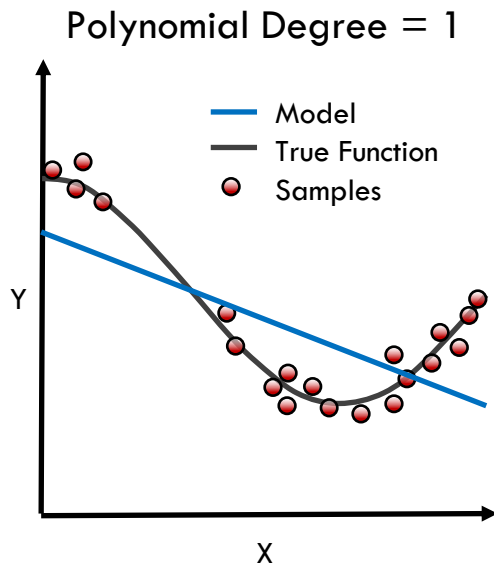


Just Right

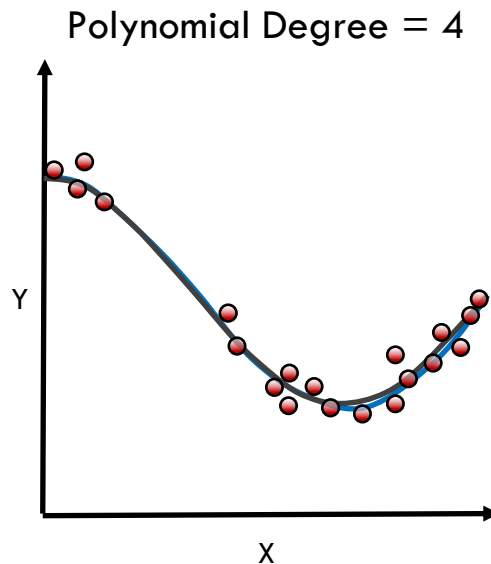


Good at Training
Poor at Predicting

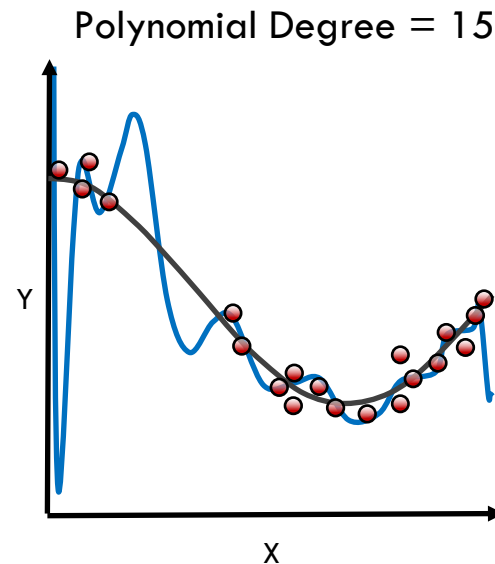
Underfitting vs Overfitting



Underfitting

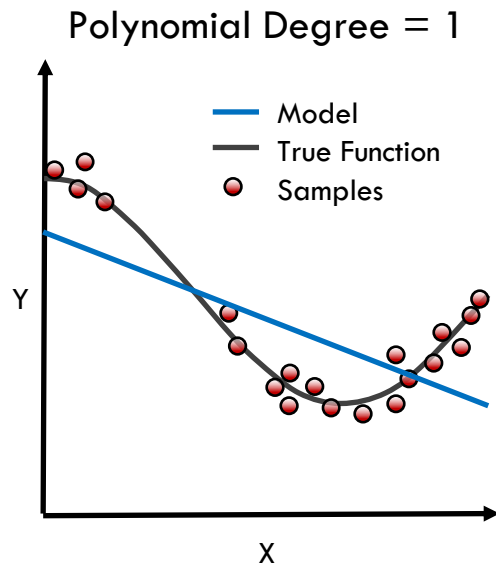


Just Right

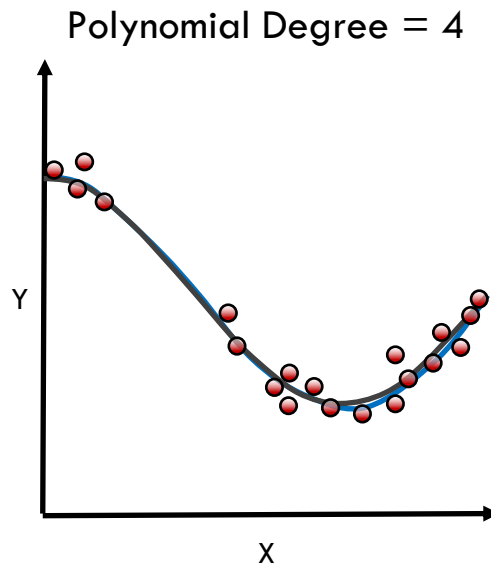


Overfitting

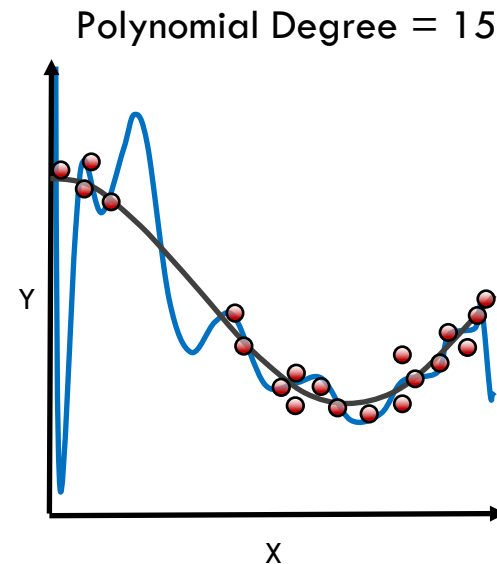
Bias – Variance Tradeoff



**High Bias
Low Variance**



Just Right



**Low Bias
High Variance**

Training and Test Splits

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5	2013-10-04	Gravity	100000000	274092705	Alfonso Cuaron	PG-13	91
6	2013-06-21	Monsters University	NaN	268492764	Dan Scanlon	G	107
7	2013-12-13	The Hobbit: The Desolation of Smaug	NaN	258366855	Peter Jackson	PG-13	161
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15	2013-08-07	We're the Millers	37000000	150394119	Rawson Marshall Thurber	R	110
16	2013-12-13	American Hustle	40000000	150117807	David O. Russell	R	138
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**Training
Data**

**Test
Data**

Using Training and Test Data

**Training
Data**

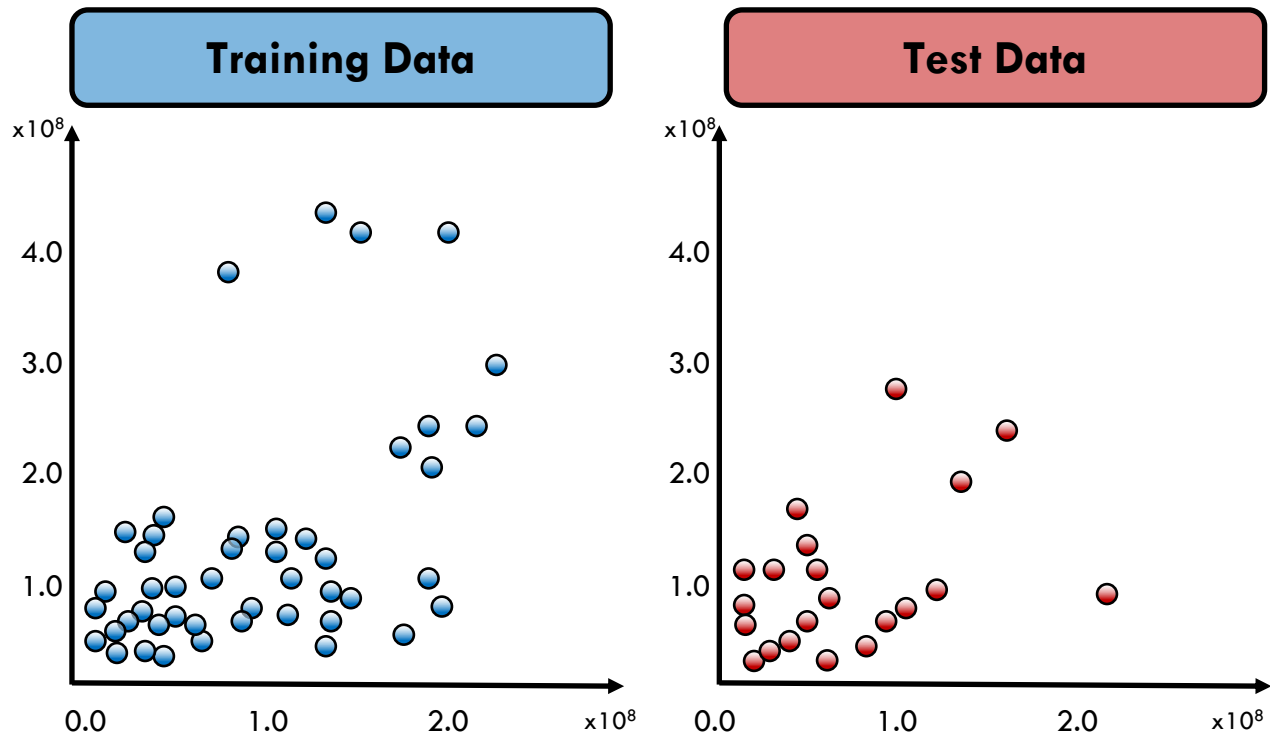
fit the model

**Test
Data**

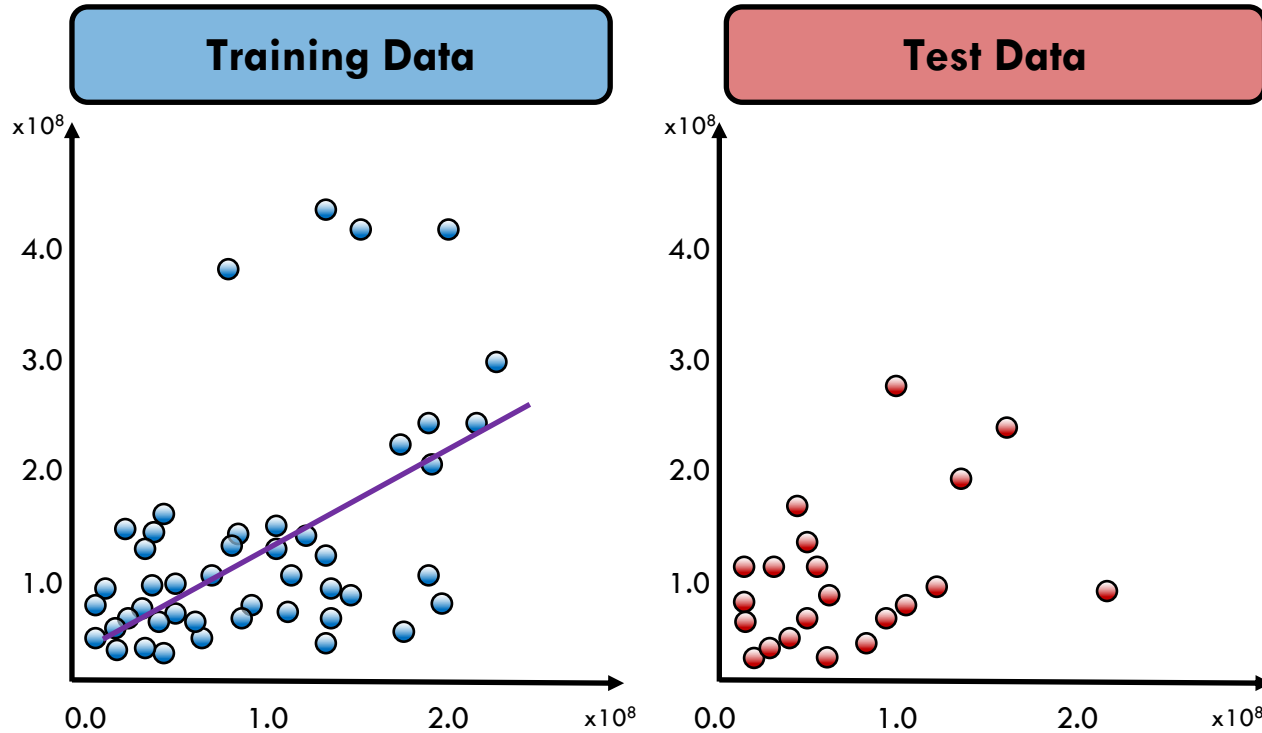
measure performance

- predict label with model
- compare with actual value
- measure error

Using Training and Test Data

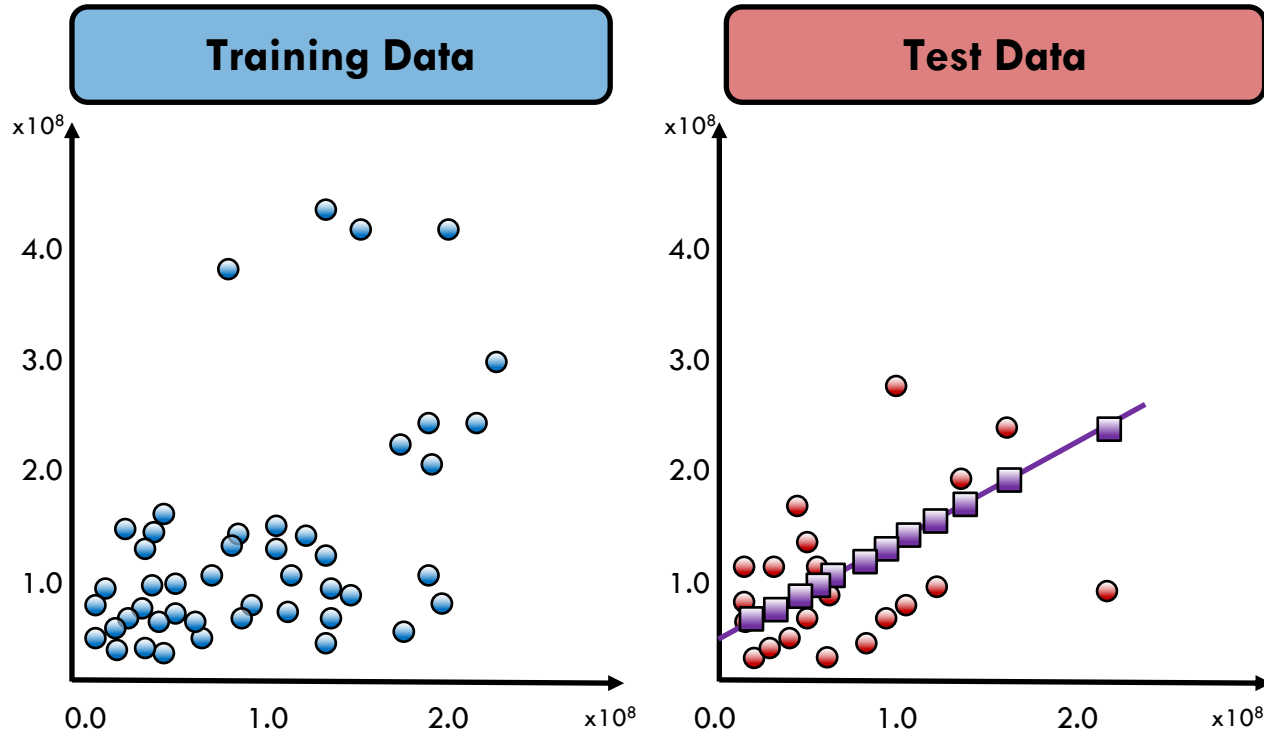


Using Training and Test Data



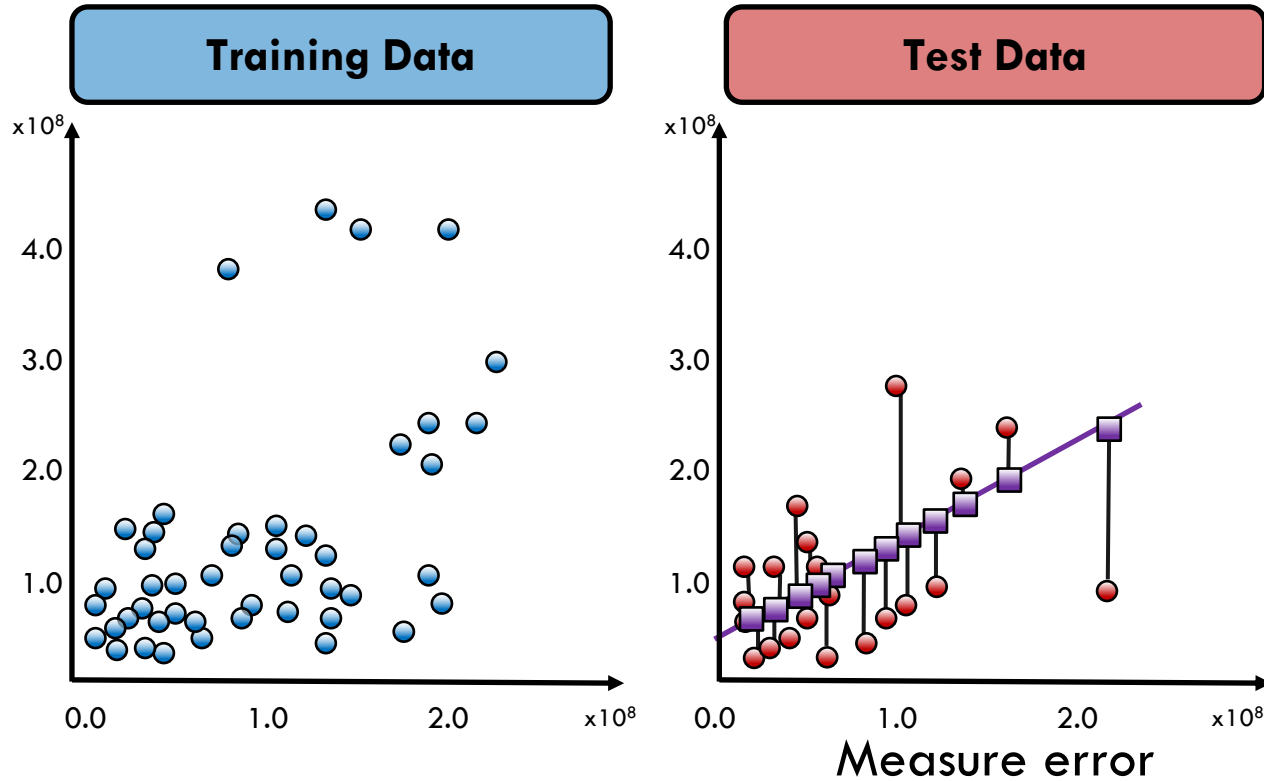
Fit the model

Using Training and Test Data

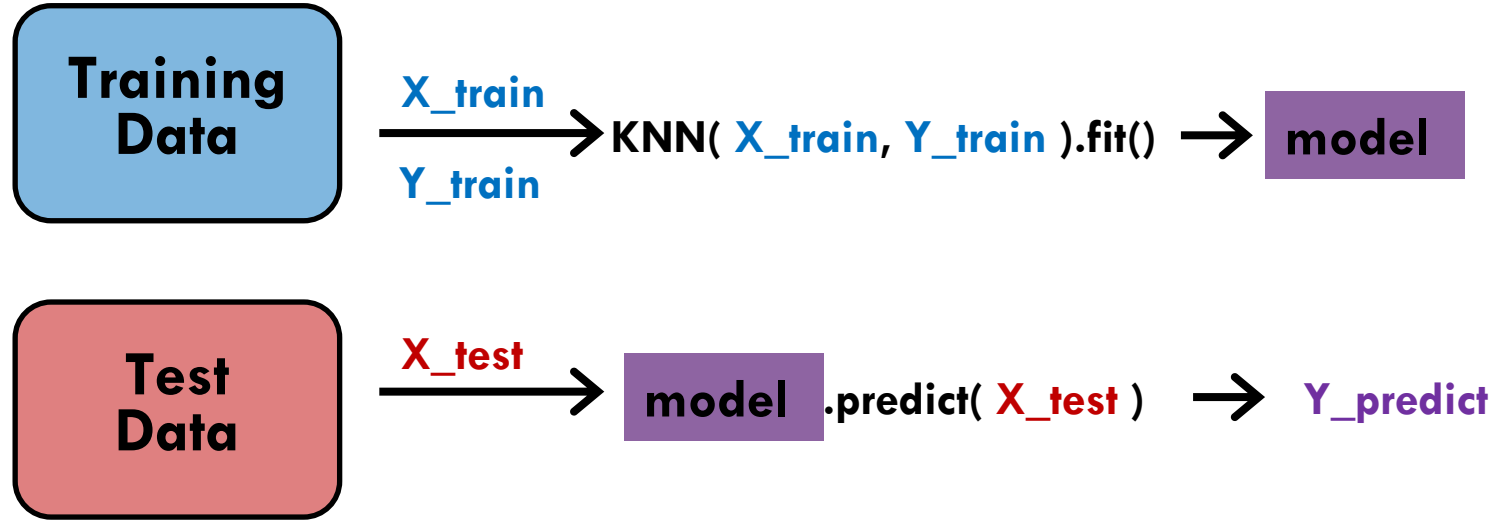


Make predictions

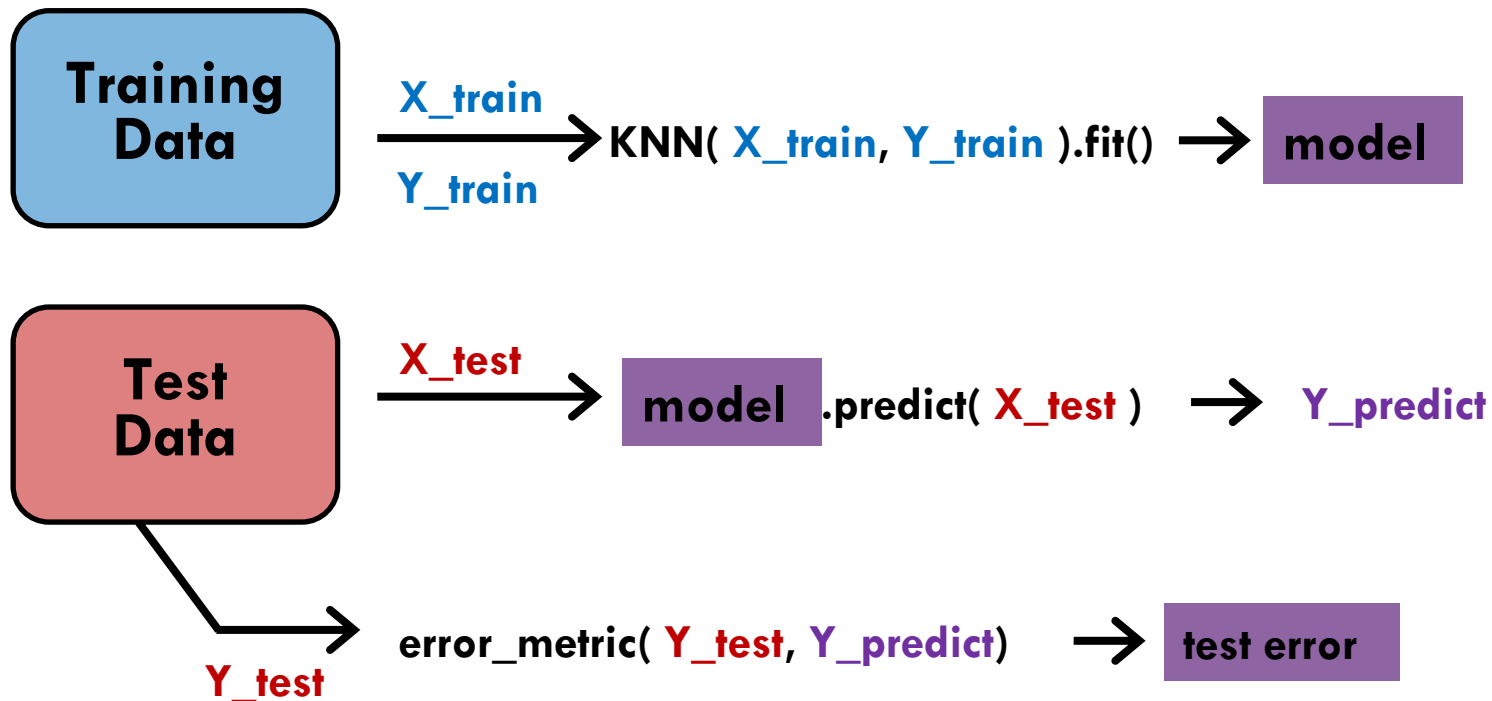
Using Training and Test Data



Fitting Training and Test Data



Fitting Training and Test Data



Train and Test Splitting: The Syntax

Import the train and test split function

```
from sklearn.model_selection import train_test_split
```

Split the data and put 30% into the test set

```
train, test = train_test_split(data, test_size=0.3)
```

Other method for splitting data:

```
from sklearn.model_selection import ShuffleSplit
```

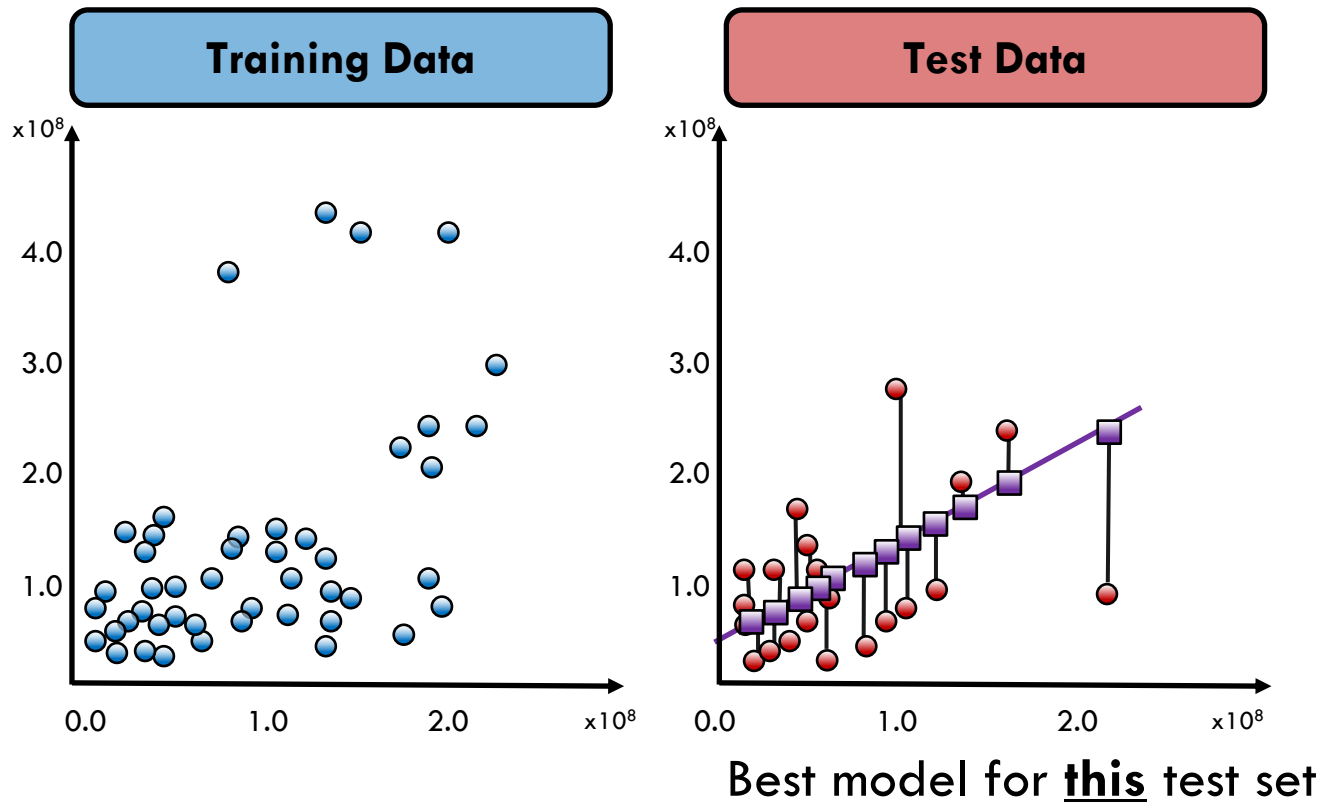
Beyond a Single Test Set: Cross Validation

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**Training
Data**

**Validation
Data**

Beyond a Single Test Set: Cross Validation



Beyond a Single Test Set: Cross Validation

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**Training
Data 1**

**Validation
Data 1**

Beyond a Single Test Set: Cross Validation

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**Training
Data 2**

**Validation
Data 2**

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**Validation
Data 3**

**Training
Data 3**

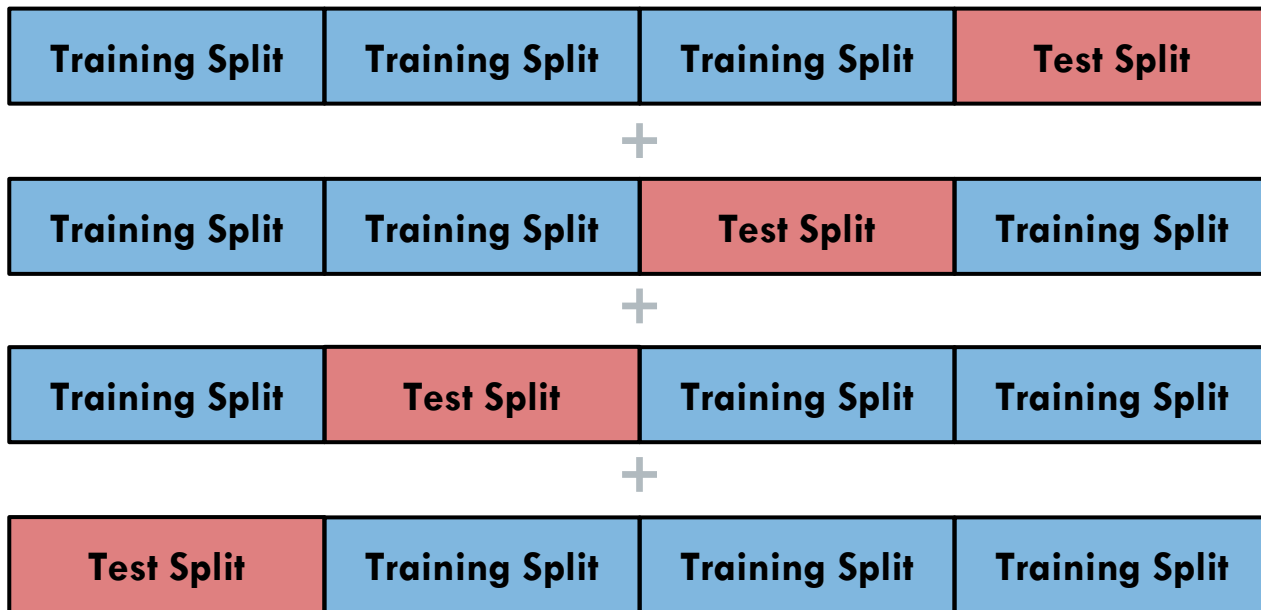
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12	2013-06-21	World War Z	190000000	202359711	Marc Forster	PG-13	116
13	2013-03-22	The Croods	135000000	187168425	Kirk De MiccoChris Sanders	PG	98
14	2013-06-28	The Heat	43000000	159582188	Paul Feig	R	117
15	2013-08-07	We're the Millers	37000000	150394119	Rawson Marshall Thurber	R	110
16	2013-12-13	American Hustle	40000000	150117807	David O. Russell	R	138
17	2013-05-10	The Great Gatsby	105000000	144840419	Baz Luhrmann	PG-13	143

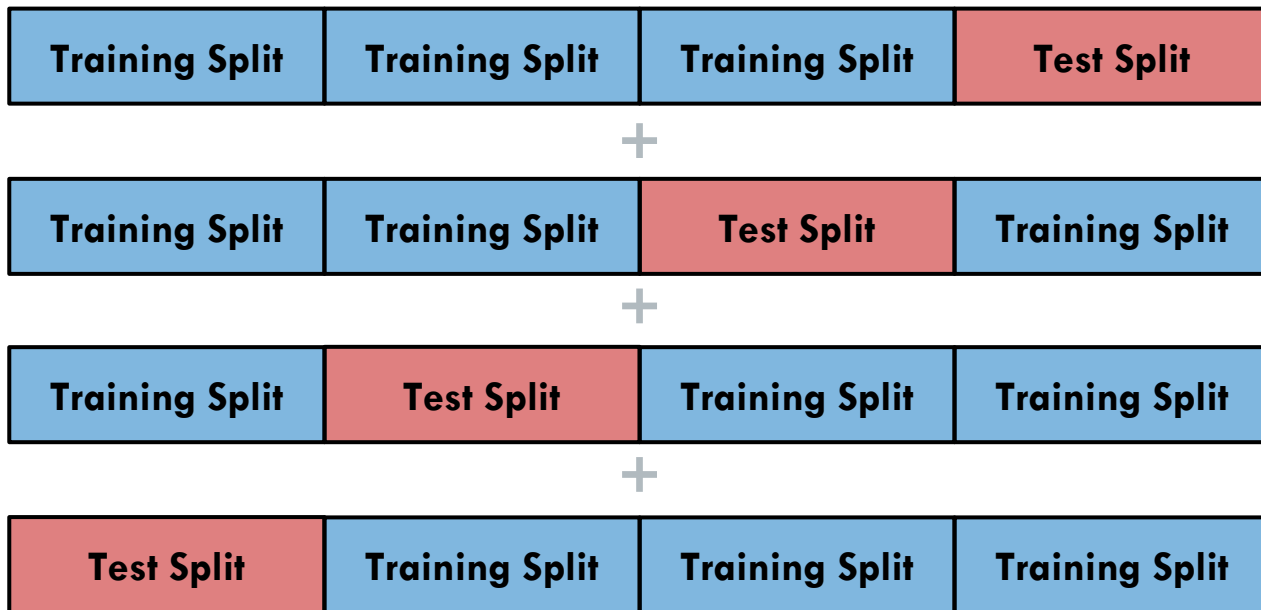
**Validation
Data 4**

**Training
Data 4**

Beyond a Single Test Set: Cross Validation

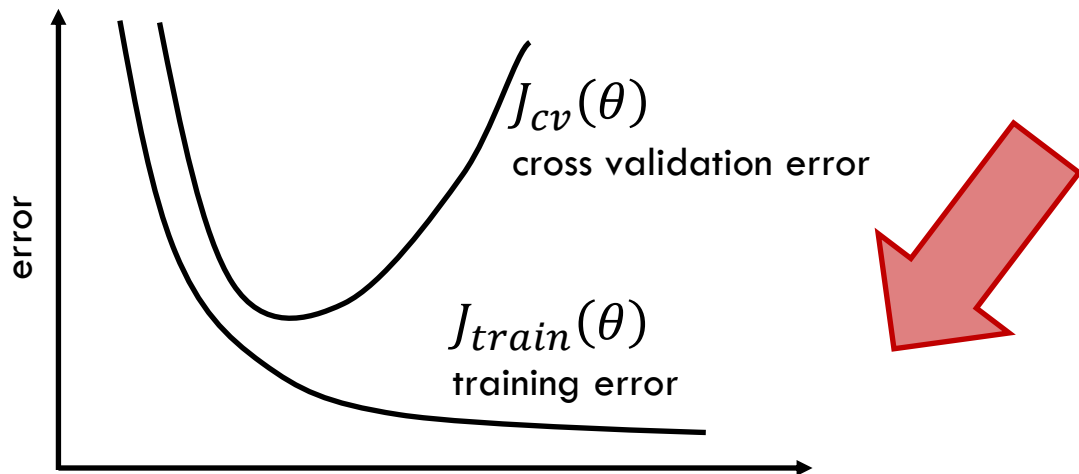


Beyond a Single Test Set: Cross Validation

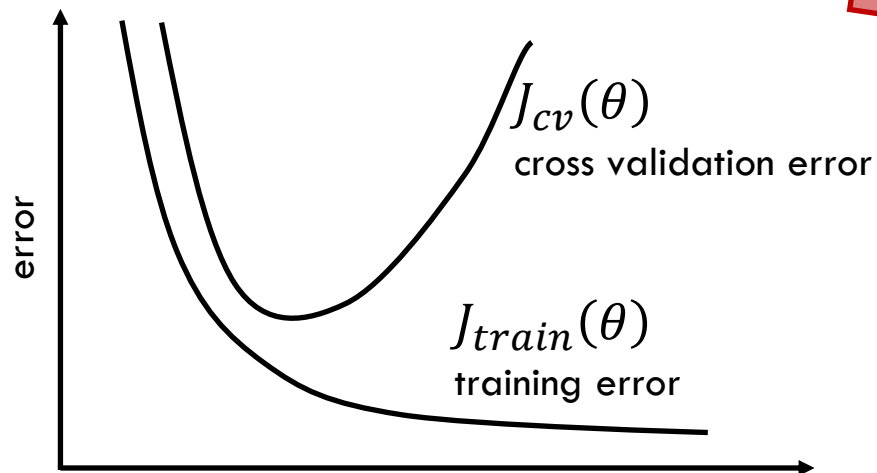


Average cross validation results.

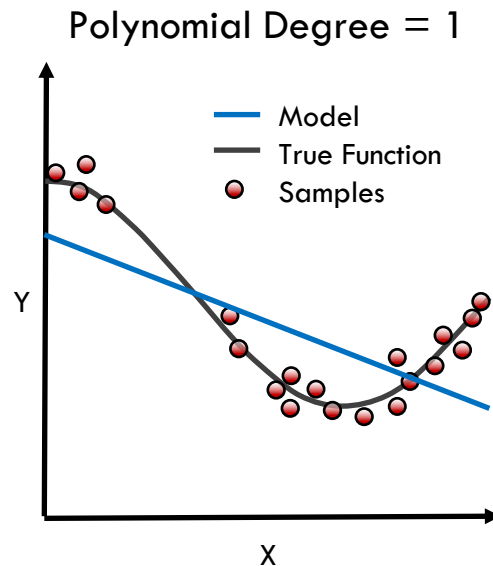
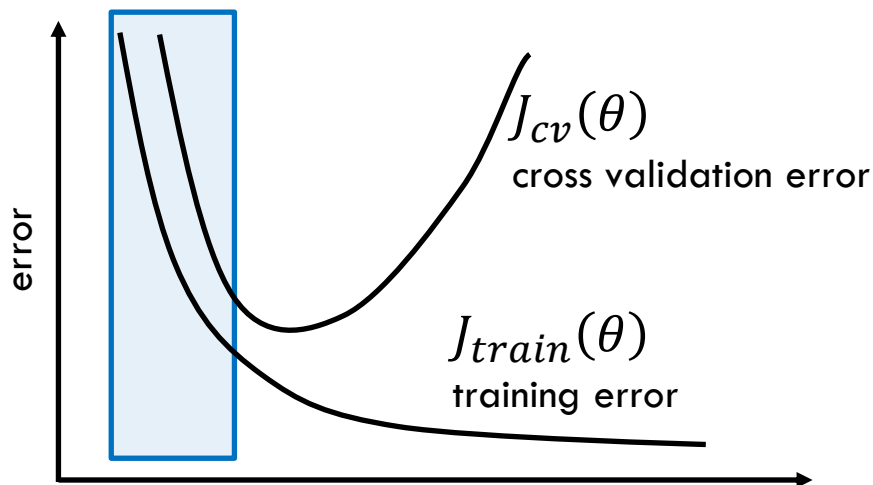
Model Complexity vs Error



Model Complexity vs Error

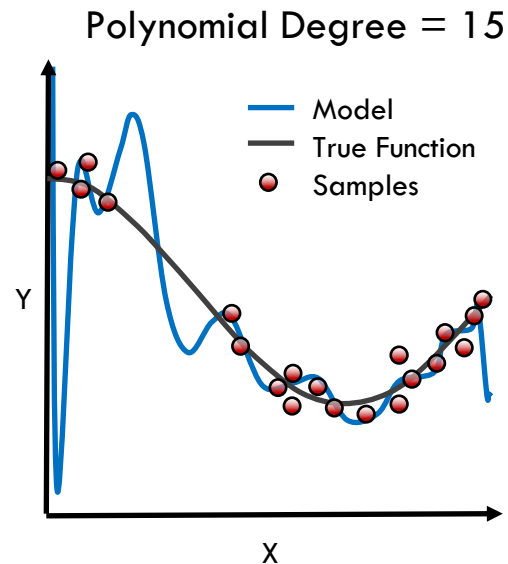
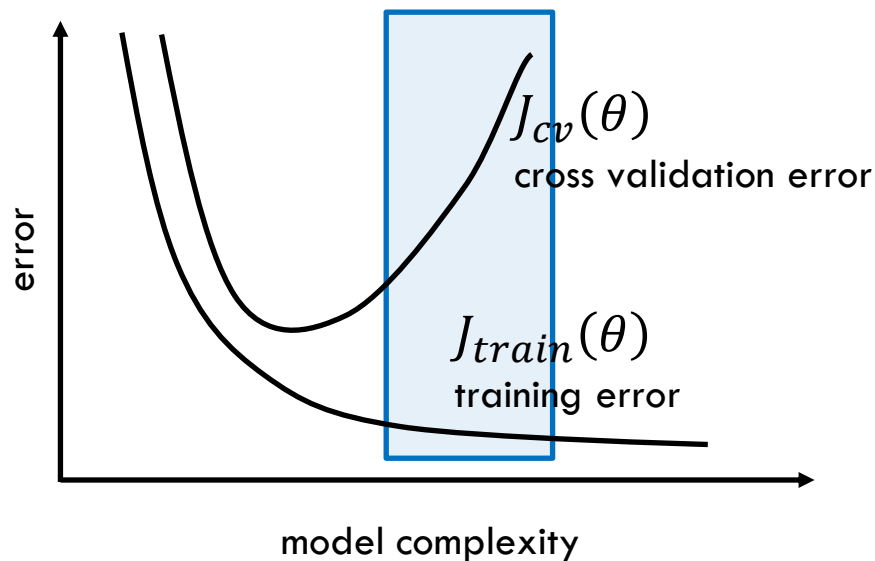


Model Complexity vs Error



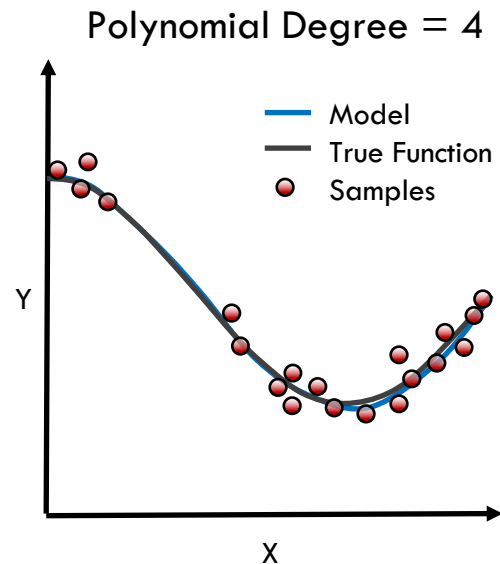
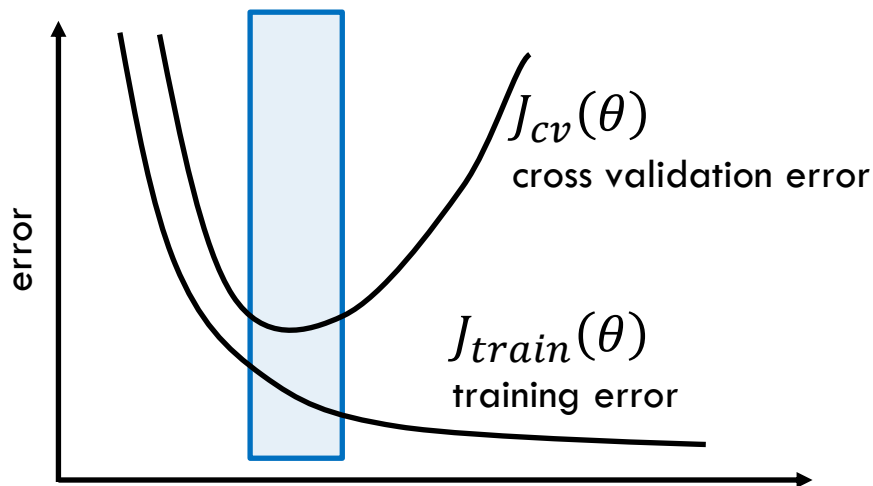
Underfitting: training and cross validation error are high

Model Complexity vs Error



Overfitting: training error is low, cross validation is high

Model Complexity vs Error



Just right: training and cross validation errors are low

Cross Validation: The Syntax

Import the train and test split function

```
from sklearn.model_selection import cross_val_score
```

Perform cross-validation with a given model

```
cross_val = cross_val_score(KNN, X_data, y_data, cv=4,  
                             scoring='neg_mean_squared_error')
```

Other methods for cross validation:

```
from sklearn.model_selection import KFold, StratifiedKFold
```